Two-Sample Tests - Lab 5.1

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May, 2021

Topics

- Two-sample test for paired data
- Two-sample test for independent group data

This lab extends inference for means in the one-sample setting to the two-sample setting. Two-sample data can consist of two samples in which the observations are paired in some way or in which the observations are drawn from two independent groups. In the paired context, inference is made about the population mean of the differences between observations; in the independent context, inference is made about the difference in population means between the two groups.

Two-sample test for paired data

Did a new wetsuit design allow for increased swim velocities during the 2000 Olympics?

Wetsuits are commonly used in the swimming stage of triathlons; in addition to providing thermal insulation against cold water, wetsuits are also thought to increase swimming speed. In 2008, de Lucas and co-authors conducted a study to assess the effect of wetsuits on swim velocity. In this study, 12 competitive swimmers were asked to swim 1,500 meters at maximal velocity, once wearing a wetsuit and once wearing a standard swimsuit. The order of wetsuit versus swimsuit was randomized for each swimmer.

The mean velocity (m/s) for each 1500m swim is recorded in the `swim` dataset.

1. Load and view the data. There are two velocity values for each swimmer: one for the wetsuit trial and one for the swimsuit trial.

a) For swimmer 1, what is the difference between velocity measured during the wetsuit trial and velocity measured during the swimsuit trial?

b) How were the values stored in the `velocity.diff` variable calculated? What does a positive value for `velocity.diff` imply, versus a negative value?[1]

c) Calculate $\overline{d}$, the mean of the observed differences.

2. Conduct a hypothesis test to address the question of interest. Let $\alpha = 0.05$.

a) Suppose the parameter δ is the population mean of the differences in velocities during a 1500m swim if all competitive swimmers recorded swim velocities with both suit types. State the null and alternative hypotheses.

b) Calculate the test statistic, where $\overline{d}$ is the mean of the differences, s_d is the standard deviation of the differences, and n is the number of differences (i.e., number of pairs). Note how the formula for the test statistic is identical to the one introduced in the previous chapter; a paired t-test is essentially a one-sample test of difference values.

$$t = \frac{\overline{d} - \delta_0}{s_d/\sqrt{n}}$$

c) Calculate the p-value from a t-distribution with degrees of freedom $n - 1$.

[1] Note: there were no negative values in these data.

d) Draw a conclusion.

3. Calculate a 95% confidence interval. Interpret the interval in the context of the data.

$$\overline{d} \pm t^\star \times \frac{s_d}{\sqrt{n}}$$

4. Verify the answers from Questions 2 and 3 using `t.test()`.

Two-sample test for independent group data

Does change in non-dominant arm strength after resistance training differ between men and women?

In the Functional polymorphisms Associated with Human Muscle Size and Strength study (FAMuSS), researchers investigated the relationship between muscle strength and genotype at a location on the ACTN3 gene. The `famuss` dataset contains a subset of the data collected from the study, which includes additional demographic and phenotypic information. The percent change in non-dominant arm strength, comparing strength after training to before training, is stored as `ndrm.ch`.

5. Load and view the data. Create a plot that shows the association between `ndrm.ch` and `sex`. Describe what you see.

6. Conduct a hypothesis test to address the question of interest. Let $\alpha = 0.05$.

a) Suppose the parameter μ_F is the population mean change in non-dominant arm strength for women, and μ_M is the population mean change in non-dominant arm strength for men. State the null and alternative hypotheses.

b) Calculate the test statistic for the difference in means of the two groups, where $\overline{x}_1$ and $\overline{x}_2$ are the means of each sample, s_1 and s_2 are the standard deviations of each sample, and n_1 and n_2 are the number of observations per sample.

$$t = \frac{(\overline{x}_1 - \overline{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

c) Calculate the p-value. The degrees of freedom for the t-statistic in this setting can be approximated as the smaller of $n_1 - 1$ and $n_2 - 1$; i.e., $\min(n_1 - 1, n_2 - 1)$.

d) Draw a conclusion.

7. Calculate a 95% confidence interval. Interpret the interval in the context of the data.

$$(\overline{x}_1 - \overline{x}_2) \pm t^\star \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

8. Verify the answers to Questions 6 and 7 using `t.test()`. Where do the values returned by `t.test()` differ from the hand calculation?